

The turbulent boundary layer as an unconfined intermittent canopy:

A structural synthesis

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Abstract

We develop a structural interpretation of the outer region of a zero-pressure-gradient turbulent boundary layer by combining three results that are usually discussed separately: the hierarchy of wall-attached momentum-transfer and vortex structures, the temporal similarity of those structures, and the geometry and dynamics of the turbulent/non-turbulent interface. The evidence supports an *intermittent-canopy* picture in which self-similar motions of many sizes overlap below the boundary-layer edge, while only the largest motions reach the intermittent region. Intense attached Reynolds-stress objects in channels have dimensions approximately proportional to their height and, at the identification threshold used in the source study, carry about 60% of the Reynolds stress while occupying less than 8% of a wall-parallel plane. Related vortex clusters have comparable aspect ratios and a scale-dependent population. Time-resolved tracking shows that tall attached structures preserve this similarity in time: their lifetimes scale with their height and u_τ . They can therefore contribute to a canopy without introducing a new outer time scale that would break wall similarity. At the outer boundary, we adopt the interface definition and conditional coordinate established by Borrell and Jiménez [2016]. Vorticity terminates there, whereas the irrotational strain outside the layer retains the potential-flow footprint of large internal motions. Moreover, strain-normalised stretching and strain-eigenvalue statistics remain similar from the turbulent core towards the interface. We hypothesise that the interface is the convoluted outer envelope of the wall-attached cluster hierarchy: the intermittent region is where that hierarchy becomes sparse. The combined evidence establishes the structural and dynamical consistency of this identification and yields direct predictions for joint cluster/interface measurements.

Keywords: wall-bounded turbulence; turbulent boundary layer; coherent structures; attached eddies; turbulent/non-turbulent interface; intermittency

1 Introduction

Descriptions of zero-pressure-gradient turbulent boundary layers (ZPG TBLs) commonly use several complementary languages. Mean-flow descriptions distinguish viscous, logarithmic, wake and intermittent regions. Structural descriptions invoke streaks, sweeps, ejections, vortex clusters and attached eddies. Interface descriptions focus on the boundary separating rotational turbulence from the irrotational free stream [Corrsin and Kistler, 1955, Borrell and Jiménez, 2016]. These descriptions refer to different observables and cannot be identified with one another without an explicit argument.

Planar particle-image velocimetry established that the outer TBL contains repeated combinations of spanwise vorticity and strong Q2 motion [Adrian et al., 2000]. Three-dimensional direct numerical simulation (DNS) subsequently showed that intense Reynolds-stress regions and vortex clusters include wall-attached, approximately self-similar families [del Álamo et al., 2006, Lozano-Durán et al., 2012]. Time-resolved DNS showed that the geometrical similarity of tall attached structures extends to their lifetimes [Lozano-Durán and Jiménez, 2014]. In

parallel, conditional analysis of the turbulent/non-turbulent (T/NT) interface showed that the rotational and potential regions are separated by a physically meaningful vorticity boundary, while the strain induced by turbulent motions extends into the free stream [Borrell and Jiménez, 2016].

We ask whether these observations can be organised into one economical picture. The proposed *intermittent-canopy hypothesis* is:

The logarithmic and outer regions contain a hierarchy of overlapping, approximately self-similar turbulent structures. Their population and spatial coverage decrease with height. Near the boundary-layer edge, the union of the surviving structures becomes sparse, and its convoluted vortical boundary is observed as the T/NT interface.

The term *canopy* conveys overlap, a distribution of heights and an irregular upper envelope. It does not imply a forest of isolated, permanent vortex tubes rooted at the wall. The structures extracted from DNS are multiscale regions rather than idealised individual vortices. In addition, most three-dimensional object evidence discussed below was obtained in channels, whereas the interface evidence was obtained in a spatially developing boundary layer. The synthesis is plausible only to the extent that the relevant logarithmic-layer organisation is shared by the two flows.

This article is a synthesis rather than a report of a new simulation. We consequently distinguish (i) observations displayed directly by published statistics, (ii) inferences supported by those observations, and (iii) the untested connection between the attached hierarchy and the interface. That distinction prevents a wall-attached Reynolds-stress object, a connected vortex cluster and a region enclosed by a T/NT interface from being treated as interchangeable.

2 Definitions and scales

Let x , y and z denote the streamwise, wall-normal and spanwise directions. The velocity fluctuations are (u, v, w) , the friction velocity is u_τ , the kinematic viscosity is ν , and δ_{99} is the height at which the mean velocity reaches 99% of its free-stream value. Wall units are denoted by a superscript $+$, so that

$$\delta_{99}^+ = \frac{u_\tau \delta_{99}}{\nu}. \quad (1)$$

A quadrant object is a connected region in which the instantaneous tangential Reynolds stress exceeds a prescribed threshold. Ejections are Q2 events ($u < 0$, $v > 0$), and sweeps are Q4 events ($u > 0$, $v < 0$). In the object analysis of Lozano-Durán et al. [2012], an object is termed attached when its lowest point lies within approximately 20 wall units of the wall. This is an operational definition of geometric connectivity, not a statement about the birthplace of the full object.

Let $\omega = |\boldsymbol{\omega}|$ be the vorticity magnitude. We adopt without modification the vorticity-interface convention and interface-conditioned coordinate established by Borrell and Jiménez [2016]. Their dimensionless interface scale is

$$\omega^* = \omega^+ \sqrt{\delta_{99}^+} = \omega \frac{\nu}{u_\tau^2} \sqrt{\frac{u_\tau \delta_{99}}{\nu}}. \quad (2)$$

The corresponding dimensional edge-vorticity scale is

$$\omega_e \sim \frac{u_\tau^2}{\nu \sqrt{\delta_{99}^+}} = \sqrt{\frac{u_\tau^3}{\nu \delta_{99}}}, \quad (3)$$

up to constants. It follows from the local-equilibrium estimate $\nu \omega_e^2 \sim u_\tau^3 / \delta_{99}$. Note that $\sqrt{\delta_{99}^+}$, rather than its inverse, appears in the definition of ω^* .

With the interface convention fixed as in [Borrell and Jiménez \[2016\]](#), let

$$\gamma(y) = \Pr\{\text{point at height } y \text{ lies on the turbulent side}\}. \quad (4)$$

This is not generally equal to $\Pr\{h_k > y\}$, where h_k is a cluster height. Pointwise coverage also depends on each object’s footprint, overlap and connectivity. A cluster-height cumulative distribution is therefore not, by itself, an intermittency profile.

3 A self-similar structural hierarchy

3.1 Three-dimensional attached objects

The vortex-cluster analysis of [del Álamo et al. \[2006\]](#) separates compact detached objects from tall attached objects. Rather than using one arbitrarily selected cluster as evidence for the hierarchy, figure 1 shows the population statistics directly. The Cartesian bounding dimensions scale approximately as

$$\Delta_x \simeq 3\Delta_y, \quad \Delta_z \simeq 1.5\Delta_y. \quad (5)$$

The data in figure 1 are from the S1900 channel at $Re_\tau \simeq 1900$, comparable to the upper friction-Reynolds-number range of the boundary-layer interface data discussed in section 4. Reynolds number is therefore not the main mismatch between the two datasets; flow geometry and object definition are. Figure 1 supplies the wall-similar cluster hierarchy required by the hypothesis; the interface side of the proposed identification is developed in section 4.

The number density per unit wall area and height has a power-law tail. As the identification threshold approaches the percolation transition, its exponent approaches -3 , as expected for a Townsend-like geometrically self-similar population [[Townsend, 1976](#), [del Álamo et al., 2006](#)].

[Lozano-Durán et al. \[2012\]](#) performed a corresponding three-dimensional analysis of intense momentum-transfer events. Their objects separate into attached and detached populations (figure 2). At their selected threshold, attached Q2 and Q4 objects occupy less than 8% of a wall-parallel plane but account for approximately 60% of the mean Reynolds shear stress at all heights. This threshold-specific statement is more defensible than claiming that such objects fill the flow or carry nearly all of the stress.

The attached Q family has dimensions

$$\Delta_x \simeq 3\Delta_y, \quad \Delta_z \simeq \Delta_y, \quad (6)$$

except for the largest objects crossing the channel centreline. Figure 3 displays the distributions underlying equation (6). The similarity between equations (5) and (6) is significant, but it does not imply that the vortex and momentum-transfer objects are identical. More than 80% of attached vortex clusters intersect a Q object, whereas the reverse probability decreases with height because tall Q objects outnumber tall clusters [[Lozano-Durán et al., 2012](#)].

The preferred arrangement is a side-by-side Q2–Q4 pair with a vortex cluster associated with the Q2 and with the shear layer below the Q4. Figure 4 compares the conditional geometry with one instantaneous realisation. The instantaneous field is appreciably less regular than its conditional average.

These observations provide three ingredients of the canopy interpretation: objects span a hierarchy of heights; their wall-parallel dimensions grow with height; and sparse intense objects make a disproportionate contribution to momentum transfer. They do not make velocity, stress and vorticity thresholds interchangeable.

3.2 The largest objects are composite

Self-similarity breaks down for the largest channel structures. Objects crossing the centreline account for roughly one third of the stress carried by attached Q objects. Fewer than 1% of

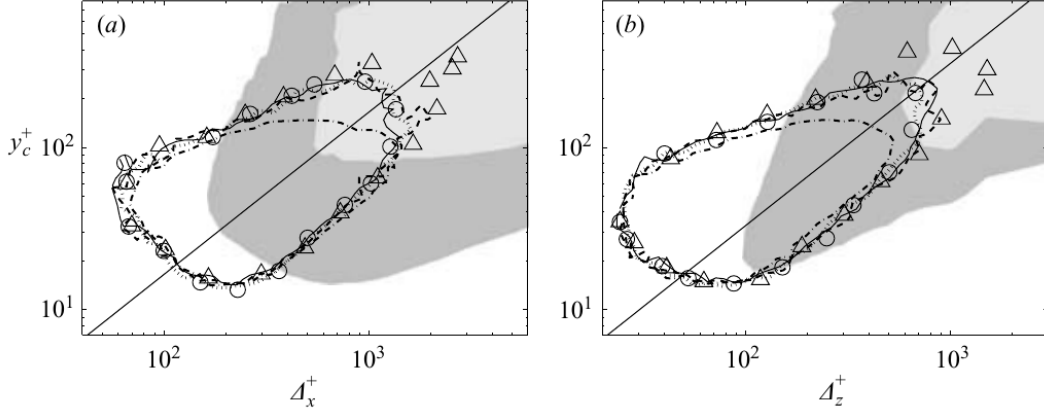


Figure 1: Joint distributions of attached-cluster streamwise (a) and spanwise (b) dimensions versus the wall distance of their centres. The diagonal trends are the direct evidence for geometrical self-similarity. Data are from the S1900 channel at $Re_\tau \simeq 1900$. Reproduced from figure 7 of del Álamo et al. [2006].

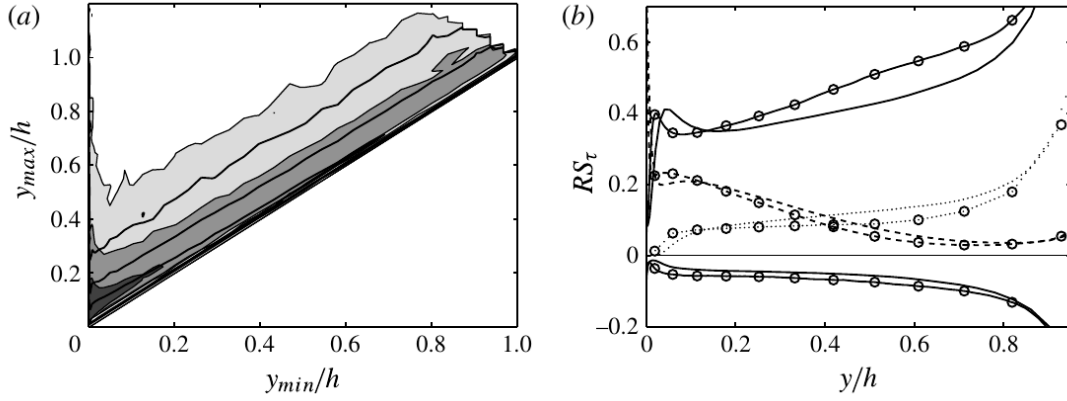


Figure 2: Left: joint distribution of the lowest and highest wall distances of identified Q objects. Right: fractional contributions to Reynolds stress from attached Q2 and Q4 objects, detached objects and counter-gradient events. Reproduced from figure 2 of Lozano-Durán et al. [2012].

all identified objects account for about 60% of the volume of the attached population, and streamwise lengths can approach 20δ [Lozano-Durán et al., 2012]. Figure 5 shows that a very-large Q2 is visibly assembled from smaller pieces.

Consequently, the height of a percolated object is not necessarily the size of one dynamical eddy, and a long correlation or spectral mode need not represent one continuous vortex packet. Spectral evidence likewise contains multiple geometrical regimes: the width of the streamwise-velocity spectrum varies approximately as the square root of its length, and the largest global modes require an outer velocity scale [del Álamo et al., 2004]. Any canopy model must distinguish active momentum-transfer motions from longer, partly inactive streamwise-velocity organisation.

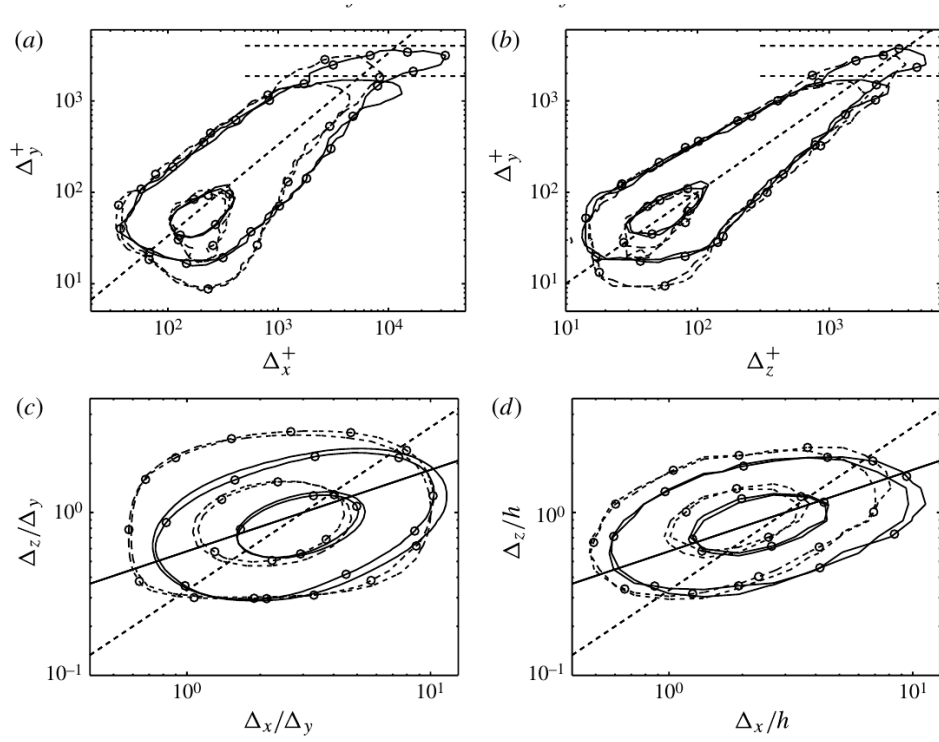


Figure 3: Joint distributions of the dimensions of attached Q2 and Q4 objects. The diagonal trends in panels (a,b) support dimensions proportional to object height. The largest objects in panel (d) depart from the simpler self-similar family. Reproduced from figure 5 of [Lozano-Durán et al. \[2012\]](#).

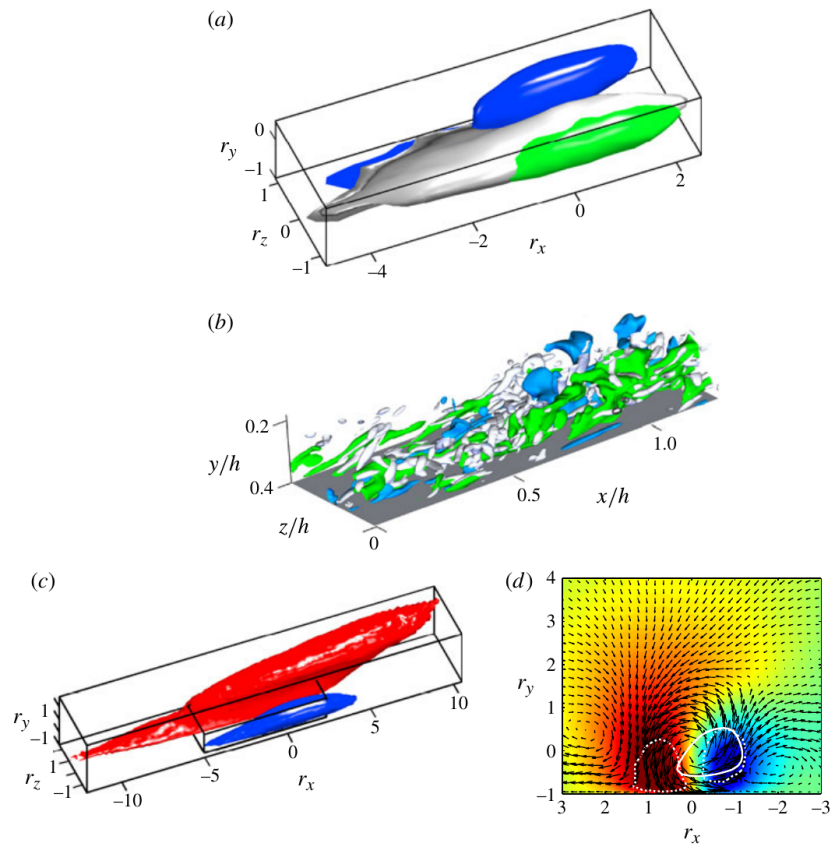


Figure 4: Conditional geometry, an instantaneous realisation, and the associated velocity field of a Q2–Q4 pair. Green denotes Q2, blue Q4 and grey the vortex cluster in the upper panels. Reproduced from figure 12 of [Lozano-Durán et al. \[2012\]](#).

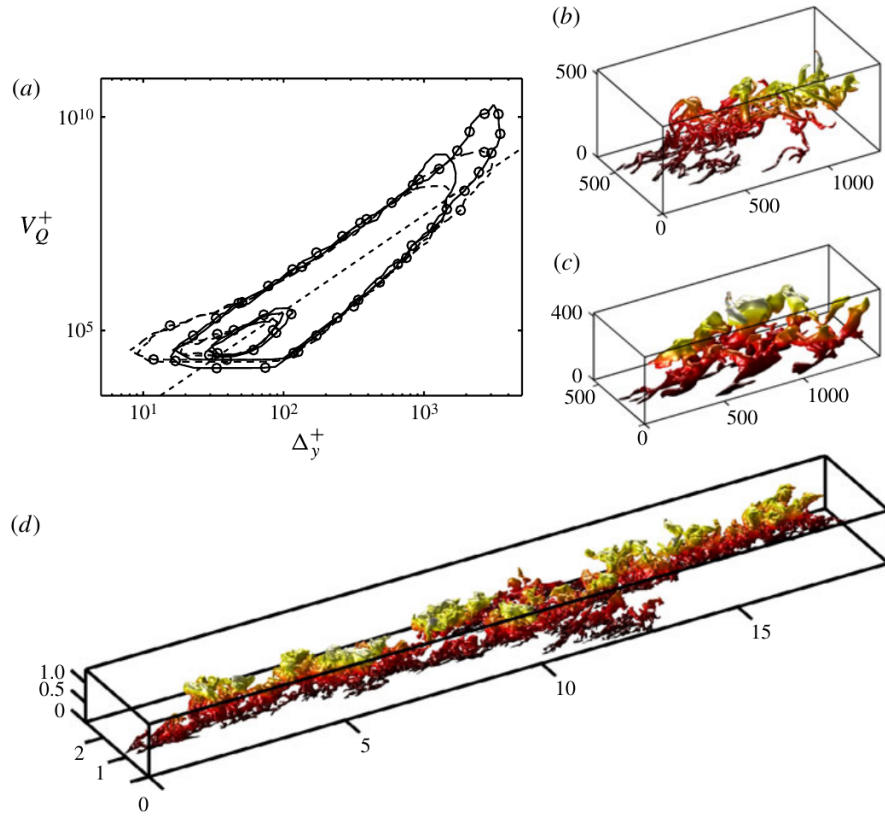


Figure 5: An attached vortex cluster (b), an attached Q2 (c) and a very-large centreline-crossing Q2 (d). The global object is visibly composite. Reproduced from figure 8 of [Lozano-Durán et al. \[2012\]](#).

3.3 Preservation of wall similarity

The time-resolved result needed by the cluster-to-interface hypothesis is whether the structures forming the canopy preserve wall similarity. [Lozano-Durán and Jiménez \[2014\]](#) found that tracked tall attached branches remain geometrically self-similar and that their lifetimes scale as

$$\frac{Tu_\tau}{\Delta_y} = O(1). \quad (7)$$

The tracked structures therefore introduce neither a separate outer length nor a separate outer time scale: their dimensions scale with Δ_y , and their duration with Δ_y/u_τ . This is the role of the time-resolved evidence in the hypothesis: the coherent structures can form a canopy while preserving wall similarity. The identification of that canopy with the boundary-layer interface is then supplied by the continuity evidence in [section 4](#).

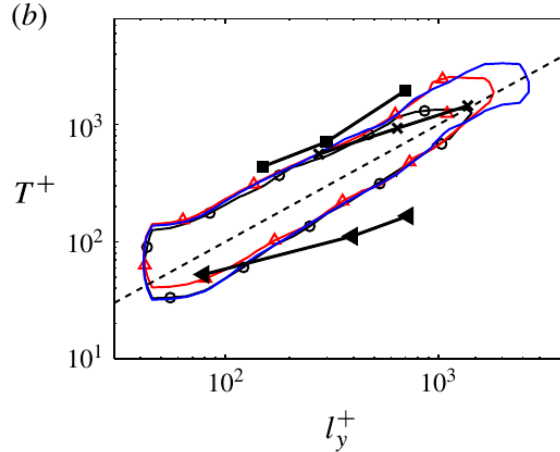


Figure 6: Lifetime of tall attached Q branches versus branch height. The dashed diagonal is $T^+ = l_y^+$. The temporal scale is set by the same height and friction velocity that describe the attached hierarchy. Cropped from figure 9(b) of [Lozano-Durán and Jiménez \[2014\]](#).

4 Continuity across the established T/NT interface

We use the interface-conditioned statistics reported by [Borrell and Jiménez \[2016\]](#) directly. The present question is what those statistics imply for the canopy framework.

4.1 The exterior strain as the shadow of internal motions

Figure 7 is a direct bridge from the turbulent field to the irrotational exterior. Vorticity falls rapidly above the boundary-layer edge, but strain persists into the free stream. Its decay follows the potential-flow behaviour associated with wall-parallel wavelengths of order $2\delta_{99}$. The exterior strain is therefore not evidence for a second vortical population. It can be interpreted as the irrotational footprint, or “shadow”, of the largest internal motions.

This result supplies a precise interpretation of the canopy metaphor. The vortical canopy has a finite, intermittent support, but the velocity and pressure fields induced by it do not terminate on that support. An observer using velocity or strain alone can therefore detect the canopy above the vorticity interface without detecting rotational turbulence there.

4.2 Core-to-interface continuity

Figure 8 provides the strongest evidence that the interface does not introduce a qualitatively separate local straining mechanism. Panel (a) follows the joint vorticity and strain distributions towards the interface, and panel (b) follows vortex stretching and compression. In panels (c,d), stretching/compression and strain eigenvalues are normalised by local strain. Their distributions at 7η and 100η from the interface are similar.

The vorticity amplitude is extinguished across the edge without a corresponding replacement of the local strain geometry. This supplies the interface side of the hypothesis: the edge inherits the internal turbulent straining environment expected if it is the outer envelope of the cluster hierarchy. Together with the wall-similar hierarchy in section 3, it motivates the cluster-to-interface identification advanced here.

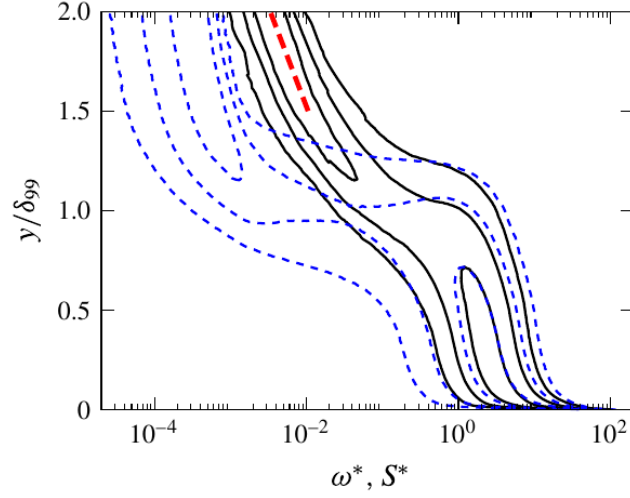


Figure 7: Joint wall-normal distributions of vorticity (blue dashed contours) and strain (black solid contours). Vorticity terminates rapidly, while strain extends into the free stream along the potential-flow decay marked in red. Reproduced from figure 20 of [Borrell and Jiménez \[2016\]](#).

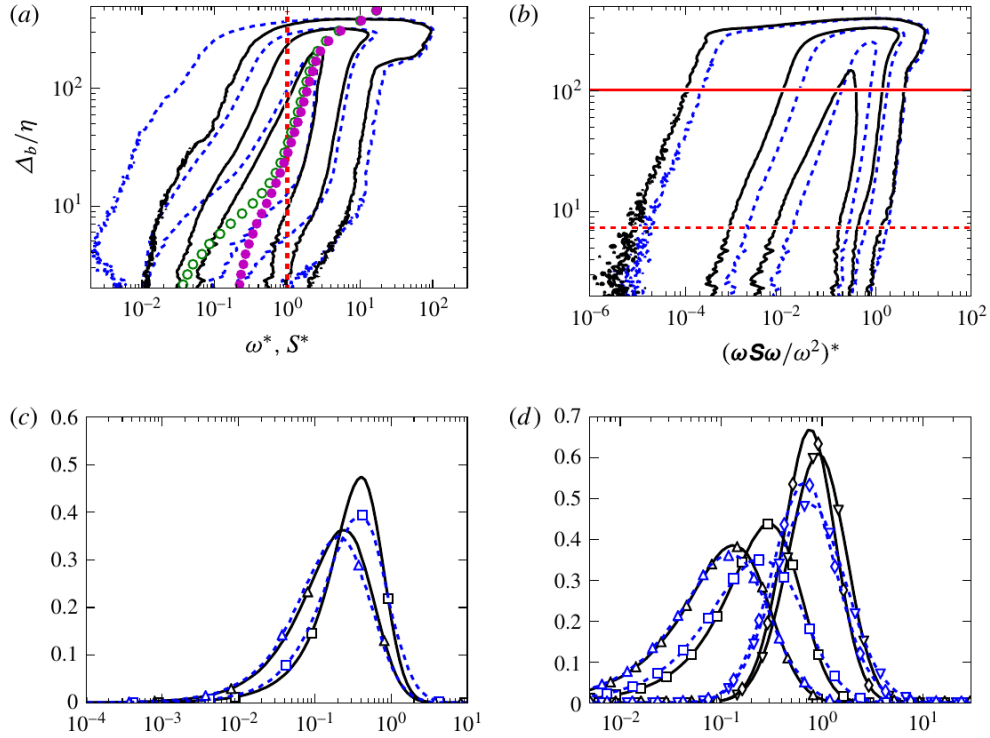


Figure 8: Conditional gradient statistics relative to the vorticity interface. Blue dashed curves in panels (c,d) are sampled at 7η and black solid curves at 100η . Their near collapse after local-strain normalisation shows that the interface retains the straining organisation of the core. Reproduced from figure 23 of [Borrell and Jiménez \[2016\]](#).

5 The cluster-to-interface hypothesis

The object and interface datasets fit together at four levels:

1. attached vortex and momentum-transfer families supply structures over a broad range of heights;
2. their dimensions grow and their population decreases with size;
3. the largest attached motions influence the full layer and the near-wall region;
4. the rotational support ends at a convoluted vorticity boundary, while the potential strain footprint and the locally normalised straining organisation remain continuous with the internal turbulence.

The central hypothesis of this paper is therefore that the intermittent region is the sparse upper support of the wall-attached hierarchy, and that the T/NT interface is its convoluted outer envelope rather than a separate sheet placed above the outer flow. This also explains why a planar section can suggest isolated bulges or packets even when the three-dimensional field is strongly connected.

A joint object/interface analysis provides a direct test of the hypothesis: vorticity clusters, Q structures and the interface should be extracted simultaneously from the same TBL snapshots and their conditional intersections measured. The observation that more than 80% of channel vortex clusters intersect a Q object supplies a concrete expectation for that test.

A similarly careful statement is required for vorticity. For incompressible, constant-density flow,

$$\frac{D\boldsymbol{\omega}}{Dt} = (\boldsymbol{\omega} \cdot \nabla) \mathbf{u} + \nu \nabla^2 \boldsymbol{\omega}. \quad (8)$$

There is no baroclinic source in equation (8). The wall supplies vorticity, stretching and tilting redistribute and amplify it, and viscosity diffuses it across the T/NT interface. This global statement does not imply a Lagrangian conveyor belt in which identifiable vortex objects travel intact from the wall to δ_{99} . The tracking evidence argues against that literal interpretation.

6 Boundary layers and channels

A channel is not a boundary layer with a reflecting outer boundary. [Jiménez et al. \[2010\]](#) show that large-scale velocity correlations cross the channel centreline; ejections from one half can penetrate the other. A TBL replaces this centreline interaction with an intermittent mixture of rotational and potential fluid and with associated pressure fluctuations. Figure 9 demonstrates centreline coupling but does not demonstrate a destructive collision between two attached hierarchies.

Higher-Reynolds-number one-point statistics show that all three velocity-component intensities exceed their channel values in the TBL, with the largest difference near $y/\delta \simeq 0.35$ – 0.5 and little movement of that location over the available Reynolds-number range [[Sillero et al., 2013](#)]. This is consistent with an edge-mediated pressure and entrainment effect, not with a dynamically empty outer TBL.

Three-dimensional correlations provide the strongest correction to the earlier “channel traffic jam” interpretation [[Sillero et al., 2014](#)]. Weak streamwise-velocity correlations reach approximately 18δ in channels but only approximately 7δ in boundary layers. Their cross-flow organisation is nevertheless qualitatively similar. Figure 10 shows that the channel streamwise correlation length continues to grow to $y/\delta \simeq 0.5$, while the boundary-layer value levels off much earlier. The source study argues that intermittency prevents the corresponding global structure from forming in the TBL.

The defensible comparison is therefore:

- a channel has no T/NT interface, communicates across its centreline and supports very long global streamwise modes;

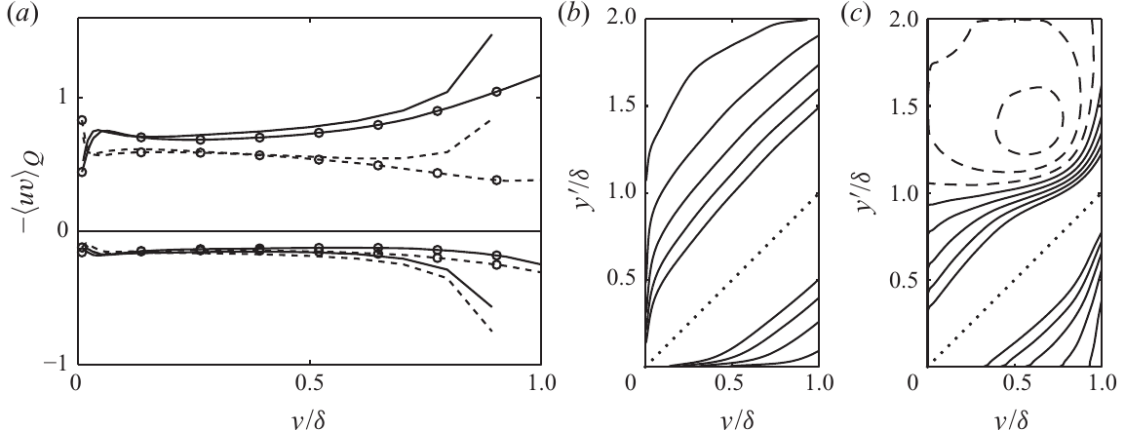


Figure 9: Quadrant contributions in a channel and a boundary layer (a), and wall-normal (b) and streamwise (c) large-scale channel correlations. Contours crossing $y/\delta = 1$ demonstrate centreline coupling. Reproduced from figure 5 of Jiménez et al. [2010].

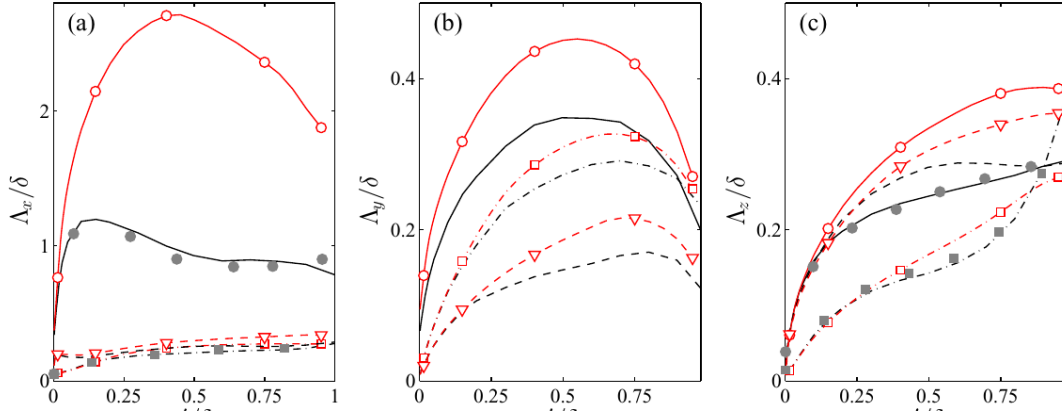


Figure 10: Integral correlation lengths for the three velocity components. In panel (a), the red channel curve for streamwise velocity continues to grow to $y/\delta \simeq 0.5$, whereas the black boundary-layer curve levels off earlier. Reproduced from figure 10 of Sillero et al. [2014].

- a boundary layer has an open intermittent edge, stronger pressure and transverse-velocity fluctuations, shorter streamwise correlations and a potential-flow footprint beyond the vorticity boundary.

“Unconfined” does not mean dynamically simpler. It means that the outer boundary condition is an irrotational free stream rather than another turbulent half-channel.

7 Direct tests of the hypothesis

The canopy interpretation can be tested quantitatively using existing DNS fields or modest extensions of them.

7.1 Joint object/interface labelling

For each snapshot, identify the T/NT interface with the established convention of Borrell and Jiménez [2016], together with attached vortex clusters and attached Q objects. Measure

$$P(d_{\text{TNTI}} < a\lambda \mid \text{top of attached object}) \quad (9)$$

and compare it with randomised objects of the same size and height. A canopy interpretation predicts an excess near the tops of the largest objects.

7.2 Coverage rather than height alone

Construct the pointwise union of objects and compare its coverage

$$\gamma_Q(y; H) = \Pr\{(x, y, z) \in \cup_k Q_k(H)\} \quad (10)$$

with the vorticity intermittency in equation (4). Object footprints and overlaps must be retained; a cumulative height distribution is insufficient.

7.3 Conditional budgets at structural tips

Evaluate mean-shear production, nonlinear transport, vortex stretching and viscous diffusion conditioned jointly on distance from the interface and distance from an object top. If sparse object tips form the interface, these budgets should differ from those at generic interface points of the same height. The test would determine whether the strain continuity in figure 8 is specifically associated with attached structures.

7.4 Matched internal/external comparison

Use identical thresholds and object definitions at matched Re_τ . Compare object populations, height scaling, centreline or interface termination and canopy coverage. This would replace qualitative collision and freedom metaphors with measurable differences.

8 Conclusions

The published evidence supports a unified but qualified structural narrative.

1. **Hierarchy.** Three-dimensional vortex clusters and intense Reynolds-stress objects contain attached, approximately self-similar families whose wall-parallel dimensions scale with their height. At the threshold of [Lozano-Durán et al. \[2012\]](#), attached Q objects carry about 60% of the Reynolds stress while covering less than 8% of a plane.
2. **Wall similarity.** Tracked tall attached structures preserve geometrical and temporal similarity: their dimensions scale with height and their lifetimes with Δ_y/u_τ . They can therefore populate a canopy without introducing a new outer scale.
3. **Core-to-edge continuity.** Vorticity terminates near the interface, but the potential strain footprint extends beyond it and the locally normalised straining organisation persists from the core towards the boundary.
4. **Internal versus external flow.** Channel structures and correlations cross the centreline, whereas TBLs interact with potential fluid through intermittency and pressure. Channels possess longer, not shorter, global streamwise correlations.
5. **Cluster-to-interface hypothesis.** The T/NT interface is proposed to be the convoluted outer envelope of the sparse wall-attached cluster hierarchy. This identification unifies the evidence above and predicts measurable joint cluster/interface intersections.

The main value of the canopy framework is not that it abolishes the outer layer. It identifies the wall-similar attached hierarchy with the structural support of the T/NT interface, identifies figures 7 and 8 as the strongest existing evidence for that connection, and turns the hypothesis into measurable predictions.

Data availability

The DNS statistics and several datasets associated with the cited studies are available through the Fluid Dynamics Group at Universidad Politécnica de Madrid: <https://torroja.dmt.upm.es/>, <https://torroja.dmt.upm.es/turbdata/> and <https://torroja.dmt.upm.es/paperdat/>. Availability differs by paper.

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